



SEAT LIGHTWEIGHTING PROJECT (MM225)

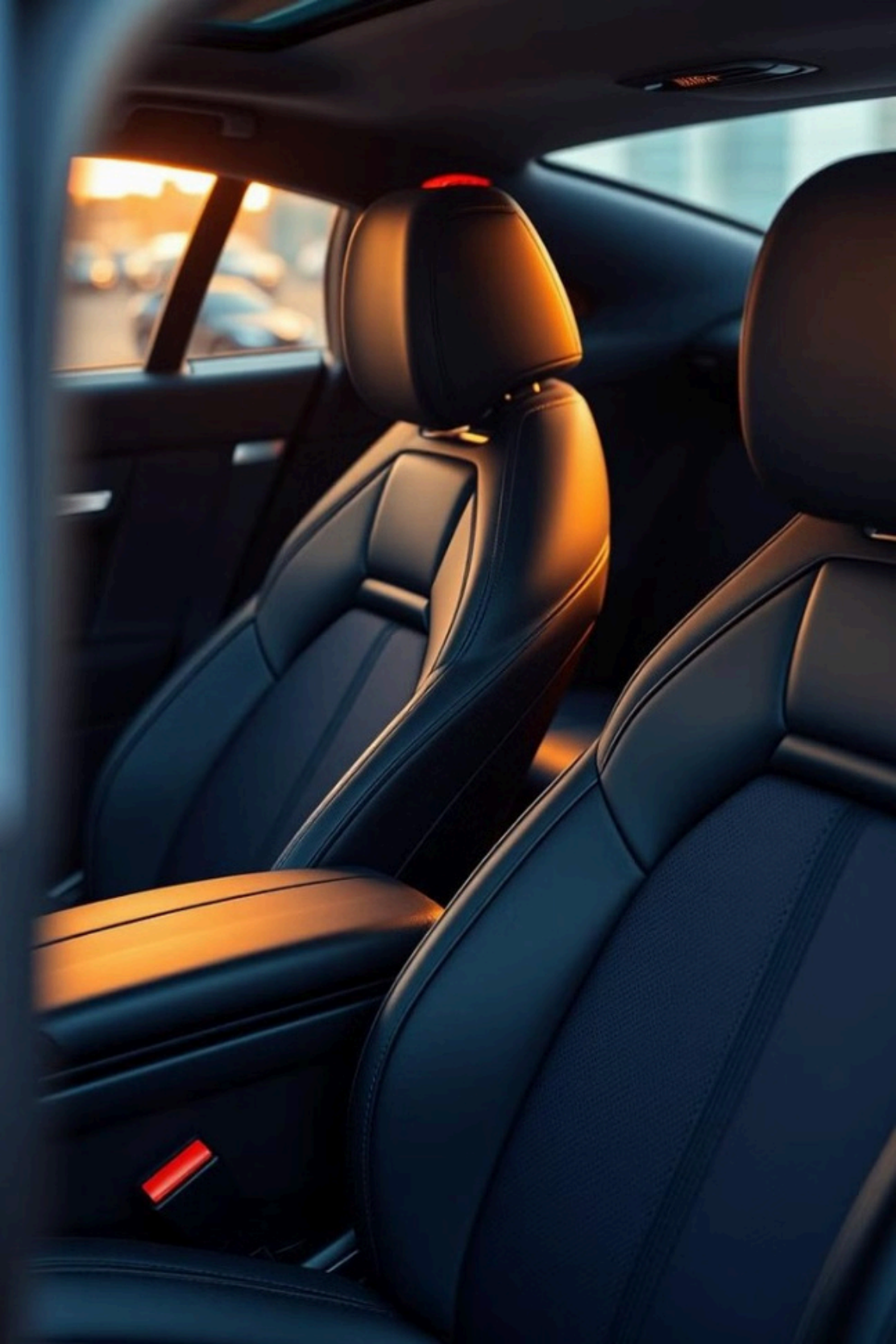
Presented to:- Dr. Hemant Borkar

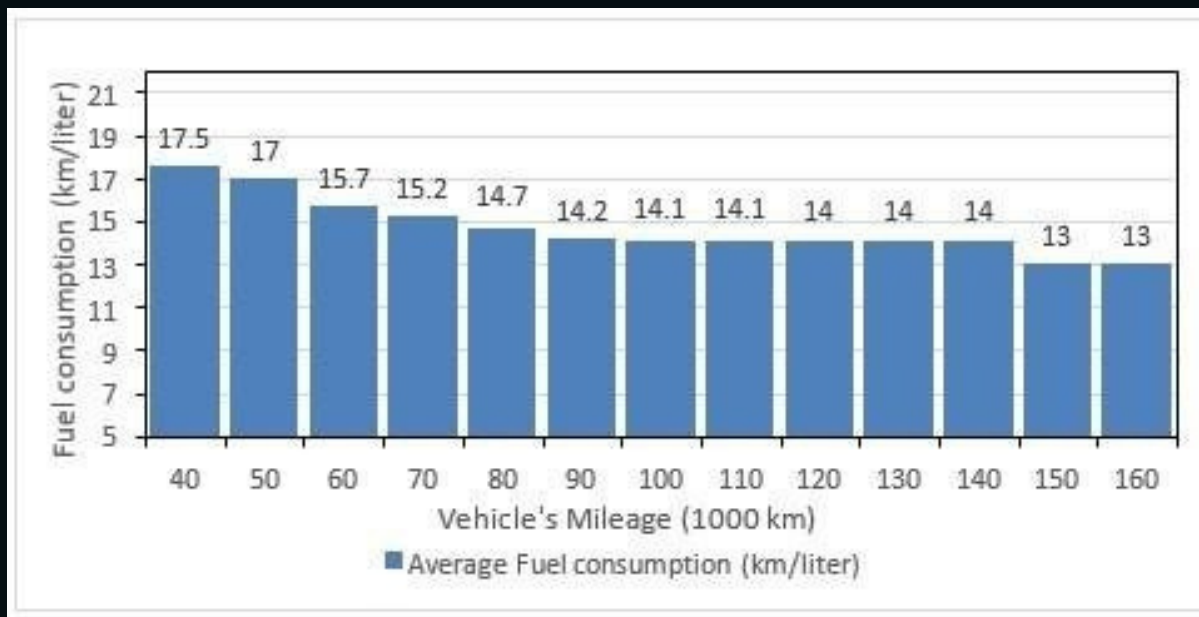
Ramavath Sunil Kumar (230005036)



Seat Weight Reduction

This presentation explores the crucial role of seat weight in automotive design and presents innovative solutions for achieving significant weight reductions. We will delve into various lightweight materials, discuss the advantages and challenges of implementing them, and highlight successful case studies.





Importance of Seat Weight in Automotive Design

1 Fuel Efficiency

Reduced weight translates into better fuel economy, lowering fuel consumption and CO2 emissions.

2 Performance

Lighter seats improve acceleration, handling, and braking performance, leading to a more responsive and enjoyable driving experience.

3 Payload Capacity

Weight reduction allows for increased payload capacity, enabling the transport of heavier cargo or passengers.

4 Sustainability

Lightweighting contributes to overall vehicle sustainability by reducing environmental impact through lower emissions.

Case Studies: Seat Weight Reduction Successes

1

Porsche 918 Spyder

Carbon fiber seats and lightweight materials contribute to its record-breaking performance and fuel efficiency.

2

Tesla Model S

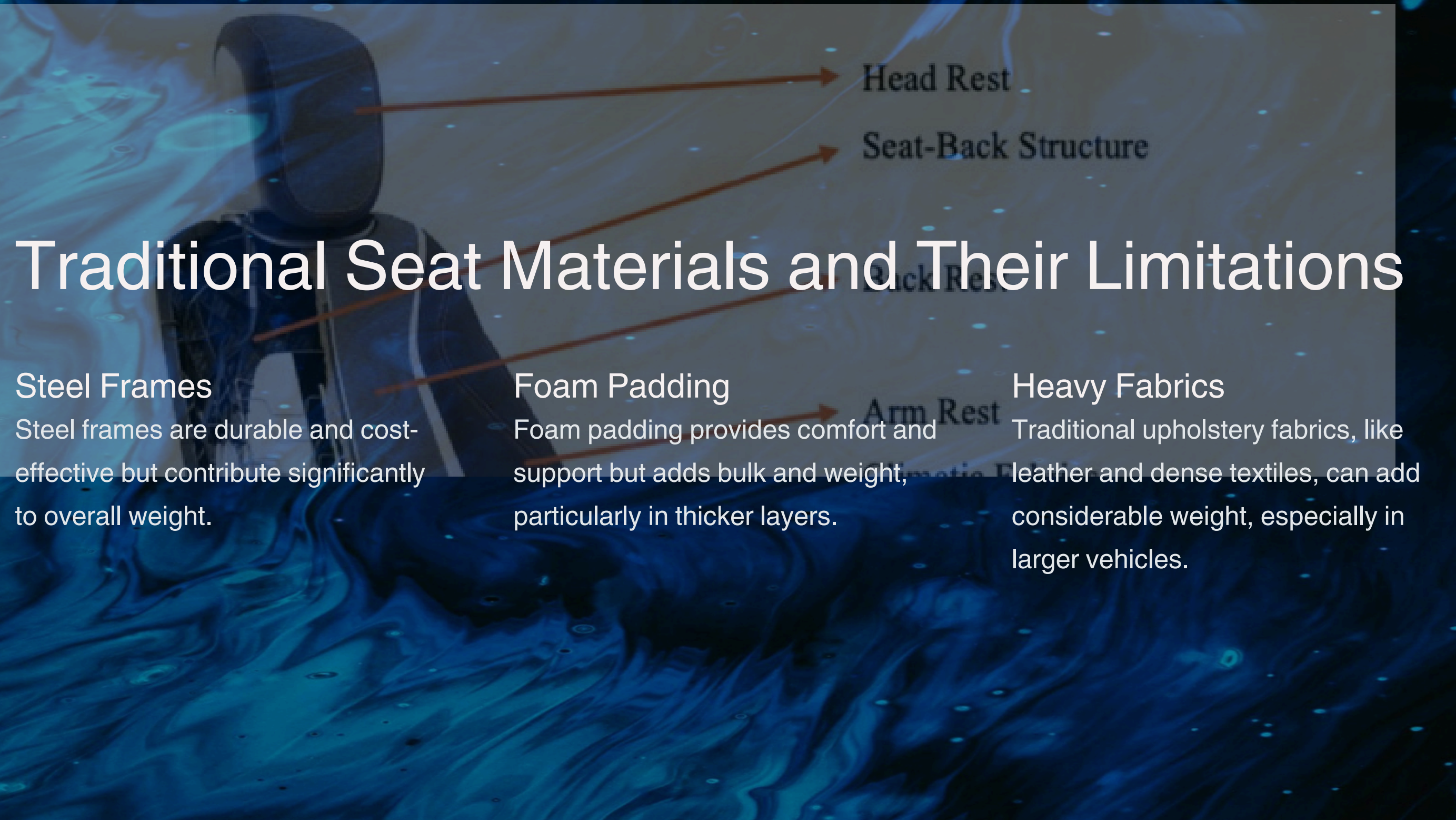
Aluminum seat frames and innovative lightweight materials play a significant role in its impressive range and acceleration.

3

Audi R8

Extensive use of lightweight materials, including aluminum and carbon fiber, results in a powerful and agile sports car.





Traditional Seat Materials and Their Limitations

Steel Frames

Steel frames are durable and cost-effective but contribute significantly to overall weight.

Foam Padding

Foam padding provides comfort and support but adds bulk and weight, particularly in thicker layers.

Heavy Fabrics

Traditional upholstery fabrics, like leather and dense textiles, can add considerable weight, especially in larger vehicles.

PARTS	ROLE	MATERIAL	WEIGHT(KG)
Base Frame Assembly	The base frame provides a stable foundation for the car seat, which helps to minimize movement during a car crash or sudden stop.	Mild Steel	5.782
Back Frame Assembly	It serves as a major foundation for driver/passenger safety, comfort and diminished fatigue	Mild Steel	4.236
Slider Assembly	Used for shifting of body weight forward or rearward	Mild Steel	3.75
Backrest Foam	Ergonomically designed and along with the soft and resilient material used provide great deal of comfort.	PU foam	1.35
Seat base Foam	Better comfort and long lasting spring back effect is necessary in selection of material.	PU foam	1.36

PARTS	ROLE	MATERIAL	WEIGHT(KG)
Hand rest Foam	Provides better ergonomically design and comfort feeling.	PU foam	0.526
Hand rest cover	Protect car armrest box from damage, spills, scuffs, dirt, pet claws and sticky fingers.	Polyester	0.044
Seat base cover	Protect the seats from the wear and tear typically associated with commuting.	Polyester	0.28
Backrest cover	Protects the back of the front seat against dirt and wear and tear.	Polyester	0.458
Headrest Foam	Provides better ergonomically design and comfort feeling.	PU foam	0.212

Advantages of Using Lightweight Materials

1

Fuel Efficiency

Reduced weight leads to better fuel economy, lowering fuel consumption and CO2 emissions.

2

Performance

Lighter cars improve acceleration, handling, and braking performance, resulting in a more responsive and enjoyable driving experience.

3

Sustainability

Lightweighting contributes to overall vehicle sustainability by reducing environmental impact through lower emissions.





Innovative Lightweight Materials for Seat Frames

Aluminum Alloys

Aluminum alloys offer excellent strength-to-weight ratios and are widely used in automotive applications, including seat frames.

Magnesium Alloys

Magnesium alloys are even lighter than aluminum, providing exceptional weight savings, though they can be more expensive and require specialized manufacturing techniques.

Carbon Fiber Composites

Carbon fiber composites offer exceptional strength and stiffness while being remarkably lightweight, making them ideal for high-performance applications.

But ,their are some more Challenges and Considerations in Material Selection.

Pugh Analysis For frame		CURRENT MATERIAL [MILD STEEL]		[XPF Steel]		[ALUMINIUM]		[Magnesium]		[CFRP]	
WEIGHTI NG FACTOR(W)	CRITERION	RATING(R) (1-10)	FUNCTIONALIT Y R x W	RATING(R) (1-10)	FUNCTIONALITY R x W	RATING(R) (1-10)	FUNCTIONALIT Y R x W	RATING(R) (1-10)	FUNCTIONALITY R x W	RATING(R) (1-10)	FUNCTIONALITY R x W
10	WEIGHT	5	50	6	60	7	70	8	80	10	100
9	RAW MATERIAL COST	7	63	6.9	62.1	5	45	4	36	1	9
8	ENVIRONME NTAL IMPACT	6	48	6	48	5	40	4	32	3	24
6	EASE OF MANUFACTU RING	3	18	4	24	6	36	5	30	2	12
8	DURABILITY	6	48	6.5	52	4	32	5	40	8	64
8	CORROSION RESISTANCE	4	32	4	32	6	48	5	40	8	64
	FUNCTIONAL ITY SUM		259		278.1		271		258		273

Based on the given criteria, the table shows that:

- XPS steel is the best alternative to replace mild steel for car seat frames.
- Aluminum has a low weight and moderate raw material cost, and it is relatively easy to manufacture.
- Magnesium has a very low weight, but the raw material cost is moderate. Manufacturing magnesium can be difficult, and it may require special equipment and processes.
- CFRP has a very low weight and very good performance, but the raw material cost is high, and it is very expensive to manufacture.

The table shows the technical and mathematical data for each material.

Carbon fiber has the highest tensile strength and elastic modulus, but is also the most expensive.

Aluminum and magnesium have a good strength-to-weight ratio and are lighter than XPS steel and mild steel, but are more expensive.

XPS Steel and mild steel have a lower cost but are heavier and less strong than carbon fiber. The final choice of material depends on the specific requirements and constraints of the project

Material	Density (g/cm ³)	Tensile Strength (MPa)	Yield Strength (MPa)	Elastic Modulus (GPa)	Hardness (HV)	Cost (/kg)
Aluminum	2.7	310-690	90-690	68.9	30-120	2-5
Magnesium	1.7	140-260	40-260	45	35-95	4-8
XPS Steel	7.23	780-920	680-820	220	120-220	0.7-2.2
Mild Steel	7.85	400-550	250-370	190-210	120-200	0.5-2
CFRP	1.5-1.6	1,000-6,000	700-3,000	230-700	--	50-200

What is xps.....?

XPS steel is a product line manufactured by Tata Steel. XPS stands for "eXtra Performance Steel," and it is designed to offer superior strength and durability for a range of applications, particularly in the automotive industry.

Its high strength and excellent levels of formability - with increased edge ductility and elongation - enable the production of robust components with complex geometry. This supports freedom in design by increasing the lightweighting potential of a part.

XPF offers weight savings of upto 30% when compared with other HSS of the same strength, and even more when replacing lower strength steels. XPF steels also provide scope for part consolidation, offering further opportunities for lightweighting and cost savings.

Substitute for
cushioning
material for back
seat ,seat base
and head rest



The claimed advantages for these "Slimline" seats were:

- no adhesives used and no hazardous emissions

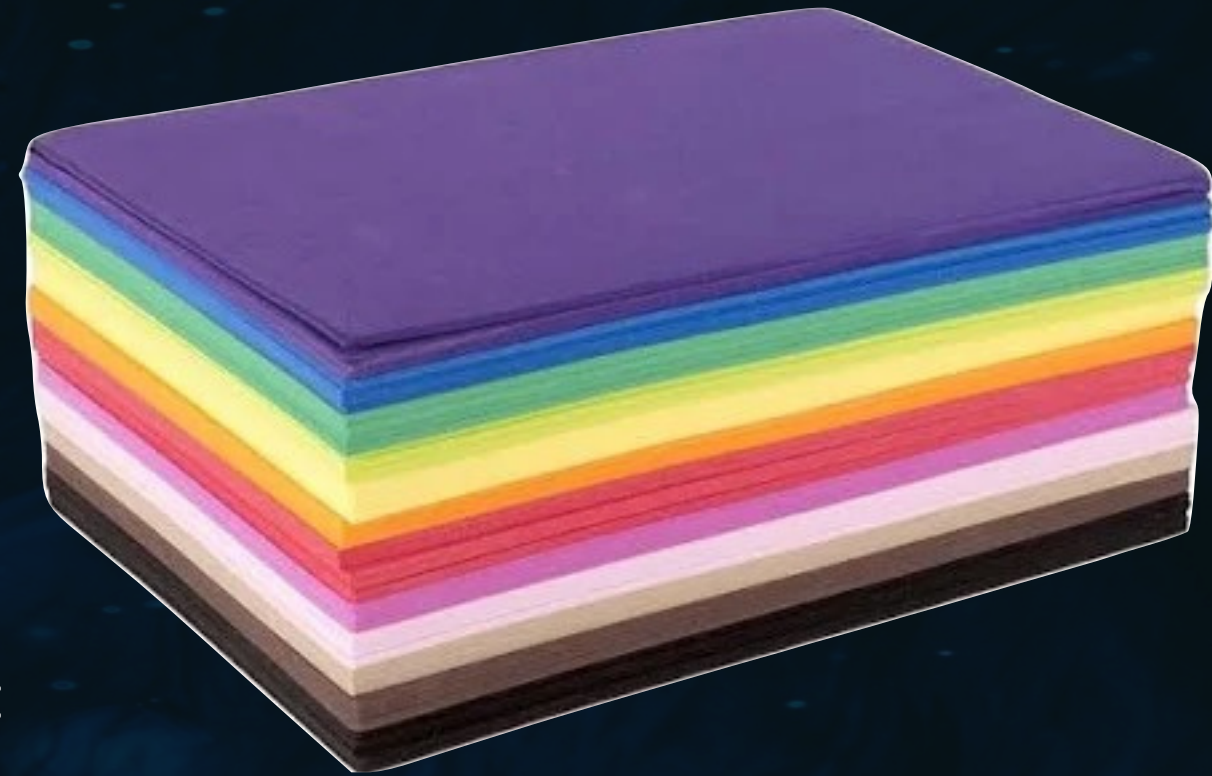
seat pads are 10-20% lighter than PU foam seats

they are up to ten times more breathable

- they can be Molded to multiple densities

- they provide equivalent load support with 10-20% weight reduction over PU foam seats

PET cushions have a lower maximum peak at resonance than PU cushioning and thus should be more "comfortable" in use (vibrational resonance testing)



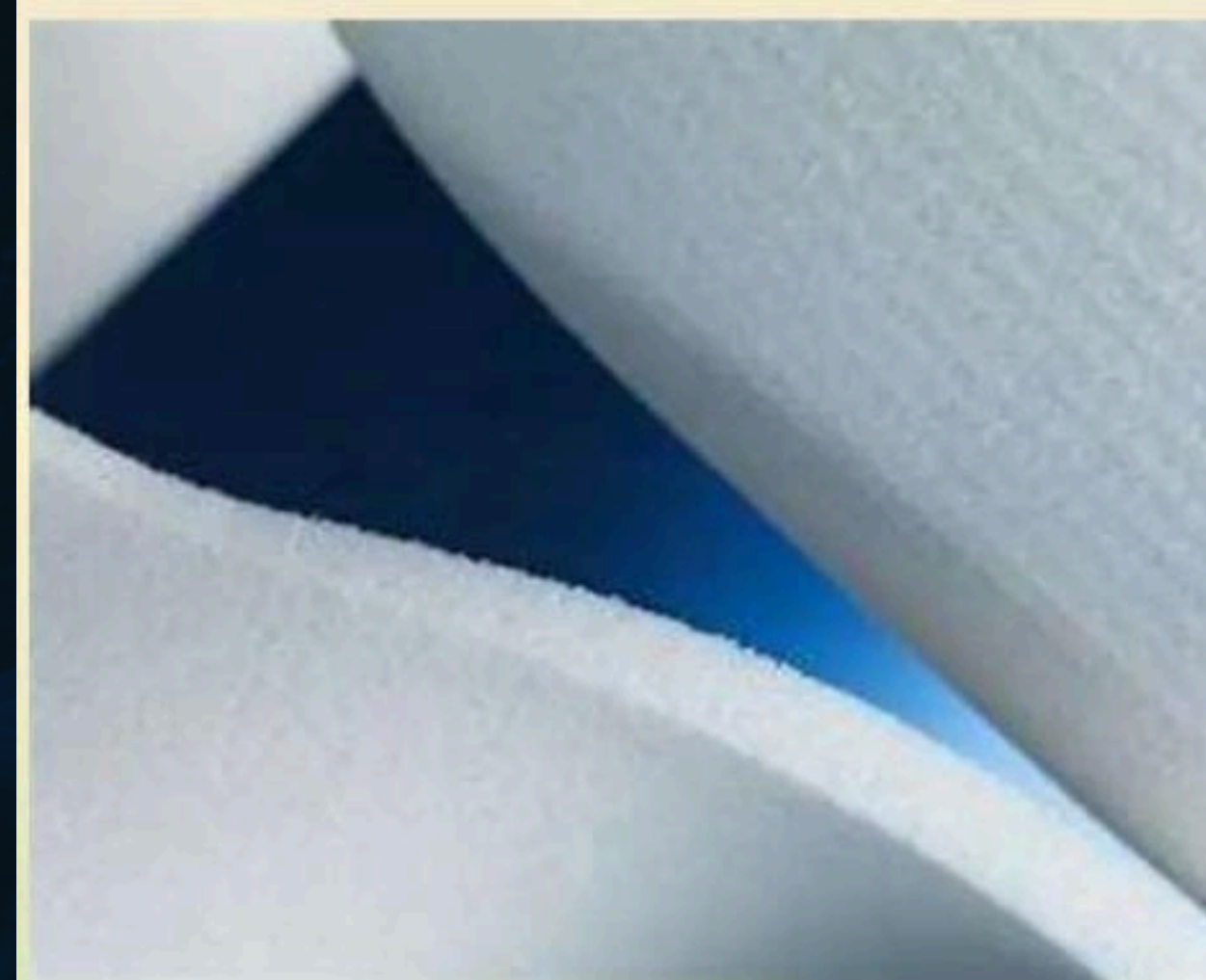
The claimed advantages for Gemstone fibric are:

- reduced weight
 - space saved
 - quick response to cabin environmental temperature change
- easy heating/cooling of seats
- potential cost savings
- * less bulk to recycle or dispose when seats are removed from expired vehicles



Ecofill are :

- 100% polyester (no Freon) and could be molded to replace PU foam seating
- molded seats were five times more breathable than PU foam seating
- the cushions will retain 85-90% of their thickness after years of use
- 30-40% lighter than PU foam
- good showroom appeal (presumably the seats "felt" good)
- recyclable



Pugh Analysis of Backrest Foam		Current material [PU foam]		New Material [Gemstone Fibre]		New Material [Slimline Fibre]		New Material [Ecofil Fibre]	
weighting factor(w)	criterion	rating(r) (1-10)	functionality r x w	rating(r) (1-10)	functionality r x w	rating(r) (1-10)	functionality r x w	rating(r) (1-10)	functionality r x w
10	weight	6	60	7	70	8	80	10	100
9	raw material cost	5	45	6	54	7	63	8	72
7	thickness retention	4	28	6	42	7	49	10	70
6	moldability	4	24	4	24	10	60	7	42
6	breathability	5	30	7	42	10	60	7.5	45
5	manufacturing cost	5	25	6	30	7	25	6	30
5	ease of manufacturing	6	30	7	35	7	25	6	30
	functionality sum		242		297		362		389

OVERALL WEIGHT REDUCTION IN BACKSEAT & SEATBASE

From 1.35kg to 0.81kg, 1.36kg to 0.816



40%



OVERALL WEIGHT REDUCTION IN (HAND REST+HEAD REST)

From 0.738kg to 0.44kg

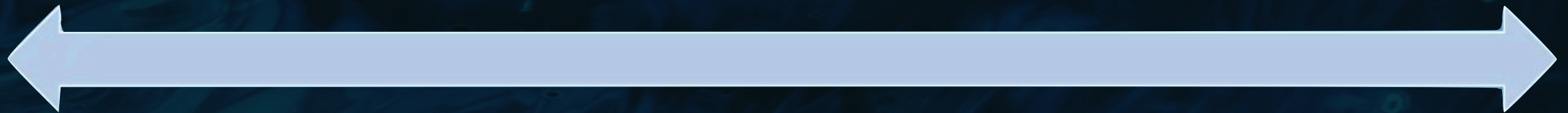


40%





SUGGESTIONS FOR POLYESTER SUBSTITUTION



Advantages and disadvantages of Nylon over Polyester

- Softer and smoother than polyester
- More abrasion resistance
- High durability than polyester
- Nylon absorbs water hence cannot dry quickly
- Gets heated in summer

<u>Pugh</u> <u>Analysis</u>		CURRENT MATERIAL [Polyester]		[Poly vinyl Fabric]		[Leather]		[Nylon Fabric]	
WEIGHTING FACTOR(W)	CRITERION	RATING(R) (1-10)	FUNCTIONALITY R x W	RATING(R) (1-10)	FUNCTIONALITY R x W	RATING(R) (1-10)	FUNCTIONALITY R x W	RATING(R) (1-10)	FUNCTIONALITY R x W
10	WEIGHT	5	50	6	60	8	80	6.5	65
9	RAW MATERIAL COST	5	45	5	45	1	9	4.5	40.5
5	MANUFACTURI NG COST	3	15	3	15	2	10	4	20
5	EASE OF MANUFACTURI NG	4	20	4	20	3	15	4	20
8	PERFORMANCE (QUALITY)	5	40	4	32	7	56	5.5	44
	FUNCTIONALIT Y SUM		170		172		180		189.5

CONCLUSION

- Total weight reduced by using XPS steel is

Part	Old Weight (kg)	New Weight (kg)
Base Frame Assembly	5.782	4.0474
Back Frame Assembly	4.236	2.965
Slider Assembly	3.75	2.625



30%

- Total weight reduced by using Ecofil fibre is

Part	Old Weight (kg)	New Weight (kg)
Back Seat	1.35	0.81
Seat Base	1.36	0.816
Hand Rest	0.738	0.44



40%

•Total Weight Reduced By using Nylon

PART	OLD WEIGHT (KG)	NEW WEIGHT (KG)
Seat base cover	0.28	0.259
Backrest cover	0.458	0.42365
Hand rest + Headrest cover	0.044	0.0407

7.5%



Conclusion and Future Trends



Sustainable Materials

Development of new materials with improved sustainability, recyclability, and lower environmental impact.



Advanced Manufacturing

Advancements in manufacturing processes, including 3D printing and additive manufacturing, leading to more efficient and customized seat designs.



Performance Optimization

Focus on optimizing the performance and functionality of seats through advanced materials and engineering.

The image features a dark blue, marbled background with swirling, organic patterns. The colors range from deep navy to a slightly lighter, tealish-blue, creating a textured, liquid-like effect. Centered in the middle of the image is the text "THANK YOU" in a clean, white, sans-serif font. The letters are evenly spaced and have a consistent thickness, giving it a modern and minimalist appearance.

THANK YOU